

Orange(Yingzhi) Ao

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ABOUT ME

Cultivated by past related experiences, my keen interest in FBSDEs and stochastic control optimization has led me to pursue a PhD, where I aim to develop innovative methodologies to enhance risk management strategies in dynamic financial environments.

References available upon request.

EDUCATION

Nanjing University B.A. in Economics - Financial Engineering Experimental Class 2021 - 2025
✓ **Overall GPA:** 92/100 **Ranking:** 3/31

✓ **Relevant Courses**

- **CS** Python, Numerical Analysis and Application Software (with MATLAB), *Financial Engineering, Applied Statistics II: Time Series (with Python), *Financial Time Series (with R), *Artificial Intelligence, *Data Mining
- **Math** Probability Theory and Mathematical Statistics, Calculus I & II, Discrete Mathematics(93, top), Linear Algebra, ODE, *PDE, *Financial Mathematics
- **Finance** Econometrics, Financial Risk Management, Securities Investment, Corporate Finance, Microeconomics, Macroeconomics, *Options Investing, Accounting(100)

Columbia University Exchange Student 2024
✓ **GPA:** 4.17/4.33 (4.0/4.0)

- ✓ **Relevant Courses** **Applied Stochastic Processes(A+), **Predictive Modeling (with R)(A), **DS in Finance And Insurance (with Python) (A+), **Insurance Pricing with ML(A+), Statistical ML (with Python) (A)

* is added before courses in progress.

** is added before grad-level courses.

RESEARCH EXPERIENCES

Multi-Period Compliance MFG-FBSDEs in REC Market RA 2024
Columbia University Stats Department

Supervised by Prof. Campbell on extending the research project "Deep Learning for Principal-Agent Mean Field Games" and its application to Renewable Energy Certificate (REC) markets

- ✓ **NN Framework With PyTorch** Set up NN models with Deep FBSDEs Solvers to validate the results of 1-period MFG studied in the previous topic.
- ✓ **Further Topics: FBSDEs For Multi-Period MFG** Extended the FBSDE Model and the NN framework to the multi-compliance case. For the next step, we will make it more realistic by adding more periods and inventory carryforward and expiration rules based on the 2-period model.
- ✓ **Innovative Numeric Tricks** Proposed and combined 3 numeric tricks to improve the numeric stability compared to the conventional Deep FBSDE Solvers, while seeking a good convergence:
 - Clamping - restrict values within $[0, 1]$ and avoid gradient explosion;
 - Sigmoid Approximation - increase differentiability and continuity
 - Logit Transformation - mapping $[0, 1]$ to $\mathbb{R}$ while providing differentiability.
- ✓ **Experimenting And Tuning** Experimented with various combos of NN settings (e.g. optimizers, schedulers, and loss functions), and fine-tuned parameters (e.g. learning rate, penalty weights), for the sake of both convergence and stability.
- ✓ **Benchmarking** Proposed Separate Optimization as a benchmark to the jointly optimized multi-period model (i.e. by running the single-period model twice), which could be interpreted as 2 distinctive perspectives: short-sighted and long-term planning.
- ✓ **Documentation, Typesetting, and GitHub-Publishing** Wrapped the models into well-structured packages and composed READMEs for code instructions. Summarized the period results into an **Overview Report**. All these works are made public in **my GitHub repository**.

Supervised by the lead researcher Yang Qu, I assisted her team in composing the research report series "Low-carbon Transition of China's Power System". Key responsibilities and contributions are as follows.

- ✓ **Co-Authorship And Editorial Support** In addition to drafting, reviewing, and proofreading the final versions of the reports, I mainly researched and co-authored two sections of the report series:
 - Design challenges and policy recommendations in the development of China's national carbon market
 - Offshore wind industry's role in the low-carbon transition - the current bottlenecks and prospects
- ✓ **Data Visualization For Whole Reports** Created all figures and tables for the report series to support data-driven insights.
- ✓ **Reference Collection** Gathered and organized references for the reports.

Machine Learning In Insurance Pricing And Risk Modeling

2024.05 - 2024.08

Columbia University Actuarial Department

- ✓ **Team Collaboration** Collaborated with a team under the supervision and mentoring of Carlos (Arocha Association) to carry out an innovative model for a data-driving insurance pricing strategy.
- ✓ **Innovative Risk Modeling For Insurance Pricing** Leveraged unsupervised learning (K-Means, PCA, etc.) to identify risk clusters based on policy data, which is novel to the current insurance industry. Developed exposure rating methods to estimate risk premiums and policy prices at each risk cluster.
- ✓ **Professional Reporting And Presenting** Delivered professional Statement of Work and Reports. Presented findings and consulting advice to clients.

PROJECT EXPERIENCES

Innovated Paired Trading Optimization with Genetic Algorithm

2023.09

Self-Motivated Personal Project

- ✓ **Innovation** Different than the traditional single-belt, I developed a double-bound, which significantly reduced fluctuations and trading turnover rate.
- ✓ **Genetic Algorithm** Self-taught and applied Genetic Algorithms (GA) to optimize key parameters such as coefficients and bandwidth, and customized fitness functions (e.g. Sharpe-ratio-based to return-based) as more appropriate model metrics.

WorldQuant Brain Consultant | Alphathon (Gold)

2023

Remote

- ✓ **Participate in NJU Alphathon 2023** Secured a gold medal (11th place out of 200+ participants) in the NJU Alphathon 2023, earning monetary rewards by submitting high-quality alphas.
- ✓ **Consultant** Contracted as a consultant to provide qualified alphas and received allowances for successful submissions

SKILLS

Programming

- ✓ **Python** Proficient in PyTorch, Pandas, Numpy, sklearn, etc., which can be supported by multiple experiences in building deep learning models and handling data science tasks.
- ✓ **R** Skilled in LM, GLM, and tree models, with practical application in portfolio management and financial time series.
- ✓ **Other languages** Familiar with C++ and MATLAB. Certified with CCF CSP-S 2nd Class. Earned CCF CSP-S 2nd Class Certificate in high school; accomplished coursework and exams with MATLAB.

Academic And Professional Typesetting

- ✓ **L^AT_EX** Used Overleaf for composing research reports, coursework, CVs, etc.
- ✓ **Professional Typesetting** Have practical experience in writing professional meeting minutes, statements of work, and presentations.

English Proficiency

- ✓ **TOEFL** 111/120 | **GRE** 333/340

Team Collaboration And Communication

- ✓ Adaptable in team settings through research projects, internships, and student associations. Skilled in providing constructive feedback and active listening.